

MIDHUN O

Software Engineer | Backend, Full-Stack & GenAI

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PROFESSIONAL SUMMARY

Software Engineer with **3+ years of experience** building backend systems, distributed processing pipelines, and AI-driven applications using **Node.js and Python**. Experienced in designing **event-driven microservices, asynchronous processing workflows, REST APIs, and cloud-based applications** for BFSI and automation platforms. Skilled in **FastAPI, RabbitMQ, Redis, AWS, LLM integrations, RAG pipelines, document intelligence, and test automation**, with full-stack experience using React.

TECHNICAL SKILLS

Backend & Languages: Python, Node.js, FastAPI, Express.js, Flask

Frontend: React.js, Next.js, ES6+ JavaScript, Tailwind CSS

Distributed Systems: Event-Driven Microservices, RabbitMQ, Asynchronous Processing, Redis, Redis/RQ

Databases & Storage: MongoDB, MySQL, AWS S3, ChromaDB, Pinecone

Generative AI: RAG Pipelines, LLM Integration, LangChain, NER, Claude, Gemini

Cloud & DevOps: AWS ECS, Docker, EC2, Lambda

Testing & Tools: Playwright Automation, Pandas, Git, Bitbucket, Jira, Postman

PROFESSIONAL EXPERIENCE

Software Engineer | MetaZ Digital, Bengaluru | Nov 2024 – Present

Test Suite AI – MetaZ (AI-Powered Test Automation Platform)

- Built a full-stack AI-powered test automation platform that **reduced manual test creation time by 80%** by converting BRDs, PDFs, and captured user journeys into structured manual and automated test assets.
- Designed and developed **FastAPI backend services** for document processing, requirement extraction, test generation, workflow management, artifact versioning, and persistence using **MongoDB and AWS S3**.
- Integrated **Claude, Gemini, and ChromaDB** into requirement-analysis, semantic-retrieval, and test-planning workflows, achieving **90% accuracy in automated Playwright script generation**.
- Built a **Chrome extension-based workflow capture system** with locator scanning and Page Object Model generation, converting recorded user interactions into reusable automation assets.
- Developed **Node.js and Playwright execution workflows** for generated automation scripts, including screenshots, execution artifacts, and HTML reports.
- Implemented **Redis/RQ asynchronous workers** to move long-running AI generation tasks outside the main API request lifecycle and support reliable background processing.

Loan Origination System (LOS) – Decimal Technologies

- Designed and developed **Node.js/Express APIs and microservices** supporting loan validation, verification, and approval workflows for BFSI clients.
- Built an **event-driven document-processing architecture using RabbitMQ and AWS S3**, enabling uploaded financial documents to be processed asynchronously outside synchronous API workflows.
- Developed **Python worker services** to consume processing jobs, retrieve documents from S3, and perform AI-driven structured data extraction from financial documents.

- Integrated external verification APIs including **GST, Udyam, and LinkedIn** to support identity and business-verification workflows within the loan-processing system.
- Developed **Python/Pandas validation pipelines** to compare AI-extracted financial information with user-submitted application data and identify discrepancies.
- Automated **First Time Report (FTR)** generation for loan assessment, reducing manual verification effort by approximately **70%** and improving the efficiency of financial-validation workflows.

Banking CAM – Decimal Technologies

- Built backend services and data-processing pipelines for analyzing large banking transaction datasets used in financial-analysis workflows.
- Implemented **LLM-based Named Entity Recognition using Qwen** to extract transaction parties, references, dates, amounts, and other structured entities from banking data.
- Developed **LLM-assisted and Pandas-based transaction classification pipelines** to identify recurring payments, self-transfers, and related-party transactions.
- Designed backend APIs for **transaction-pattern analysis and anomaly detection**, supporting financial risk-assessment workflows.
- Optimized transaction-processing workflows, reducing analysis time by **40%** for large banking statements and complex transaction histories.

Software Engineer | Coddle Technologies, Kozhikode | Jun 2023 – Oct 2024

Greenplus – CRM Platform

- Developed **Node.js/Express backend APIs** for lead management and customer onboarding
- Built backend services integrated with **React applications** to support end-to-end customer onboarding.
- Implemented automated **PDF generation workflows** using submitted form and onboarding data.
- Integrated **Yousign e-signature APIs** to support digital document execution and verification.
- Implemented customer communication and authentication workflows using **SendGrid email automation**.

RAG-Based AI Chatbot for E-Commerce

- Built backend services for a **Retrieval-Augmented Generation customer-support chatbot**.
- Developed document-ingestion and embedding pipelines using **LangChain**.
- Stored vector embeddings in **Pinecone** to enable semantic retrieval across product documentation and FAQs.
- Integrated LLMs to generate **context-aware responses grounded in retrieved product information**.
- Developed APIs enabling real-time chatbot interaction with frontend applications.

EDUCATION

Bachelor of Computer Science

University of Calicut | March 2023

CERTIFICATION

Develop Generative AI Applications: Get Started — IBM (Coursera) | February 2026

- Covered LangChain fundamentals, prompt engineering, chains, and agents for building Generative AI applications.
- Built a Generative AI web application using Flask and structured LLM outputs.